

t-Tests in R

PS50008A Week 14 | Lab Walkthrough

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5 February 2026

Welcome to Lab

Today's Lab

- Use an interactive app to explore t-tests
- Understand the difference between **independent** and **paired** t-tests
- See how means, SDs, and sample size affect your results
- Learn why we report **both** p-values and effect sizes

The App



Launch the t-test Explorer

Open the app now: [Choose your own t-test adventure!](#)

Keep it open throughout the lab!

You'll see:

- **Left panel:** Controls for your experiment
- **Right panel:** Data, distribution, and results

Part 1: Two Types of Design

Which t-test do you need?

The Key Question

! Before choosing a test, ask:

Are the **SAME** people or **DIFFERENT** people in each condition?

- Different people → **Independent samples t-test**
- Same people → **Paired samples t-test**

Between-Subjects Design

Different people in each group

Characteristics:

- Two separate groups
- Each person in ONE condition only
- Compare Group A vs Group B

Examples:

- Treatment vs Control
- Lab A vs Lab B
- Method A vs Method B

→ Use **Independent samples t-test**

Within-Subjects Design

Same people measured twice

Characteristics:

- One group of participants
- Each person in BOTH conditions
- Compare Time 1 vs Time 2

Examples:

- Before vs After treatment
- Monday vs Friday performance
- With vs Without distraction

→ Use **Paired samples t-test**

Quick Check

A researcher measures stress before AND after a meditation session (same people).

Which design?

A. Between-subjects

B. Within-subjects

Activity: Part 1

Open the activity page and complete **Part 1:**
Understanding the Two Designs

Answer questions 1-4

04:59

Part 2: Your First Experiment

Between-subjects design

Setting Up the Experiment

In the app, set:

Setting	Value
Design	Between-groups
DV name	Memory Score
IV name	Word Type
Level A	Abstract
Level B	Concrete
Level A mean	50
Level B mean	58
Both SDs	10

What to Look For

Distribution tab:

- Two overlapping curves
- Blue = Abstract, Orange = Concrete
- How much overlap?

t-test Results tab:

- t-value and p-value
- Cohen's d (effect size)
- SIGNIFICANT or NOT SIGNIFICANT?

Demo

Let me show you...

[Live demo in the app]

Activity: Part 2

Complete **Part 2: Your First Experiment**

- Set up the experiment as shown
- Run the analysis
- Answer questions 5-8

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Part 3: Changing the Means

What happens when groups differ more or less?

Experiment: Means Far Apart

Change Level B mean from **58** to **70**

What happens?

- Curves separate more
- p-value gets smaller
- Cohen's d gets larger

Experiment: Means Close Together

Change Level B mean to **51** (close to Level A's 50)

What happens?

- Curves overlap almost completely
- p-value likely > 0.05 (not significant)
- Cohen's d is tiny

Activity: Part 3

Complete **Part 3: Experiment - Changing the Means**

- Try means = 70 (far apart)
- Try means = 51 (close together)
- Answer questions 9-11

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Part 4: Changing Variability

What happens when data is more or less spread out?

Reset First!

Set means back to: **50** and **58**

Experiment: High Variability

Change both SDs from **10** to **25**

What happens?

- Curves are wider
- More overlap
- Harder to detect the difference
- Cohen's d is smaller

Experiment: Low Variability

Change both SDs to 5

What happens?

- Curves are narrow
- Barely overlap
- Difference is very clear
- Cohen's d is large

The Insight

! Signal vs Noise

- **Signal** = difference between means
- **Noise** = variability (SD)

More noise makes the signal harder to detect!

Activity: Part 4

Complete **Part 4: Experiment - Changing the Variability**

- Try SD = 25 (more noise)
- Try SD = 5 (less noise)
- Answer questions 12-14

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Part 5: Changing Sample Size

The crucial difference between p and d

Reset First!

Set back to: means = **50** and **58**, SDs = **10**

Experiment: Small Sample

Change N from **30** to **10**

What happens?

- p-value gets larger (harder to reach significance)
- Cohen's d stays about the same!

Experiment: Large Sample

Change N from **10** to **100**

What happens?

- p-value gets smaller (easier to reach significance)
- Cohen's d stays about the same!

The Key Insight

! Why This Matters

p-value depends on sample size

Cohen's d does NOT depend on sample size

With enough participants, even a TINY effect becomes “significant”!

This is why we ALWAYS report effect sizes.

Activity: Part 5

Complete **Part 5: Experiment - Changing Sample Size**

- Try $N = 10$ (small)
- Try $N = 100$ (large)
- Answer questions 15-17

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Part 6: Within-Subjects Design

Same people, measured twice

Setting Up a Within-Subjects Study

In the app, set:

Setting	Value
Design	Within-groups
DV name	Mood Rating
IV name	Time
Level A	Before
Level B	After
Level A mean	5
Level B mean	7
Both SDs	2

What's Different?

Look at the results:

- Test is now **Paired samples t-test**
- Effect size is **d_z** (for paired data)
- Interpretation talks about “change” not “group difference”
- Story hook says “twice for each participant”

Activity: Part 6

Complete **Part 6: Within-Subjects Design**

- Set up the within-subjects experiment
- Run the analysis
- Answer questions 18-20

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Part 7: Design Your Own

Get creative!

Your Challenge

Design TWO studies:

Study 1 (Between-subjects):

- Think of two groups to compare
- Name your variables
- Set realistic parameters
- Run it!

Study 2 (Within-subjects):

- Think of a before/after scenario
- Name your variables

Activity: Part 7

Complete **Part 7: Design Your Own Study**

Be creative with your study ideas!

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Summary

Choosing the Right Test

Design	People	Test
Between-subjects	Different in each group	Independent t-test
Within-subjects	Same, measured twice	Paired t-test

What Affects Significance?

Factor	Effect on p-value
Larger mean difference	Smaller p (easier)
Smaller SDs	Smaller p (easier)
Larger sample	Smaller p (easier)

What Affects Effect Size?

Factor	Effect on Cohen's d
Larger mean difference	Larger d
Smaller SDs	Larger d
Sample size	NO EFFECT

Why Report Both?

! The Complete Picture

- **p-value** tells us IF the effect is real
- **Effect size** tells us HOW BIG it is

A significant result isn't always a meaningful result!

Final Activity

Complete the **Final Knowledge Check** (questions 21-25)

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Key Takeaways

! Today We Learned

1. **Independent t-test:** Different people, `t.test(x, y)`
2. **Paired t-test:** Same people, `t.test(x, y, paired = TRUE)`
3. **$p < .05$:** Reject null hypothesis
4. **Always report:** t, df, p, d, and 95% CI

Next Steps

1. Complete the **Week 14 Self-Assessment Quiz**
2. Review **APA reporting format** for assessments
3. Prepare for **Week 15: Correlation** - relationships between variables

Assessment Reminder



Research Plan Due Week 16

Start thinking about your hypotheses and planned analyses. t-tests are likely relevant for your project.

Questions?